Status	Finished
Started	Thursday, 29 October 2026, 7:30 PM
Completed	Thursday, 29 October 2026, 7:33 PM
Duration	3 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Decreases the efficiency of algorithms
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Sorting the entire list in one pass
- b. Bubbling up the largest element to its correct position
- c. Dividing the list into halves
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Bubbling up the largest element to its correct position with each pass

b. Sorting data in a non-sequential manner

c. Using the divide-and-conquer approach

d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items

b. Sorting helps in finding duplicates quickly

c. Sorting is rarely used in programming

d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **6**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By breaking the input into smaller parts and merging them
- c. By comparing elements sequentially
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Sorting
- c. Searching
- d. Merging

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Selection
- d. Extraction

The correct answer is:
Insertion

Question **9**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It does not require recursion
- b. It is based on the divide-and-conquer approach
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Combining sorted sublists
- b. Sorting the entire list sequentially
- c. Dividing the input into smaller subproblems
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It only works on small datasets
c. It avoids the need for recursion
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **14**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Sorting
c. Searching
d. Rearranging

The correct answer is: Sorting

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. All of the above
c. Searching
d. Selection

The correct answer is: All of the above